

RMM System -- Complete Handover and User Guide

Remote Monitoring and Management Platform

Prepared for: All staff -- new joiners, IT support, technicians, administrators, and developers

Foreword

This document is the definitive reference for everyone who uses, operates, or maintains the RMM system. It is written to serve several audiences at once: the junior staff member who has never used an RMM tool before, the experienced technician who needs a quick reference, the administrator who manages users and automation, and the developer who extends or maintains the codebase.

Read the section that applies to your role. If you are unsure which sections apply to you, consult the "How to Use This Book" chapter immediately following the Table of Contents.

This guide reflects the actual codebase as implemented. All step-by-step instructions refer to real UI elements, real field names, and real system behavior. Where a feature is queued for a future phase, this is noted explicitly so you are not confused when you cannot find it.

Table of Contents

How to Use This Book

PART I -- GETTING STARTED (for everyone)

PART II -- DAILY OPERATIONS (beginner-friendly)

PART III -- ADVANCED OPERATIONS (technicians and admins)

PART IV -- TECHNICAL REFERENCE (developers and admins)

Appendix A: Glossary

How to Use This Book

This book is structured for different readers. You do not need to read it cover to cover. Use the table below to find your relevant chapters.

Your Role	Essential Reading	Also Useful
-----------	-------------------	-------------

Callout boxes are used throughout this document:

NOTE: Extra context or background information.

TIP: A faster or smarter way to do something.

WARNING: Something that can cause problems if done incorrectly.

IMPORTANT: Information you must not skip.

PART I -- GETTING STARTED

Chapter 1: What is the RMM System?

What it is

RMM stands for Remote Monitoring and Management. An RMM system is a platform used by IT teams to monitor, manage, and support computers and servers -- often for multiple clients -- from a single central dashboard.

Think of it like a control tower for an airport. Every aircraft (your managed devices) sends constant status updates. The tower (the RMM) watches all of them, raises alarms when something is wrong, and allows controllers (your IT staff) to take action remotely.

This system is modeled after industry tools like NinjaOne and ConnectWise Automate but built specifically for your organization. It is composed of four main components:

1. The Dashboard (Streamlit frontend)
2. The API (Flask backend)
3. The Agent (Python)

The agent also receives commands from the API -- such as "run this script" or "reboot" -- and executes them.

4. The Database (PostgreSQL)

Who uses it

| Role | What they primarily do |

How it fits your workflow

When a client's machine has a problem -- say the hard drive is nearly full -- the following chain of events occurs automatically:

1. The agent on that machine reports disk usage of 92% in its heartbeat.

This entire process can happen without any manual intervention up to step 6 -- that is the power of an RMM system.

Chapter 2: Logging In and Navigation

What it is

The login page is the entry point to the system. You need a valid email address and password to access any part of the platform.

Who uses it

Everyone.

Step-by-step: Logging In

1. Open your web browser (Chrome, Firefox, or Edge are recommended).

NOTE: The system uses JWT tokens for authentication. Your session token is stored in browser memory only -- it is not visible in the URL. If you copy and share a page URL, it will not contain your session credentials. To share access, the other person must log in with their own credentials.

WARNING: There is no "forgot password" button in the current version. If you are locked out, contact your system administrator to have your password reset directly in the database.

Understanding the Interface Layout

After logging in, the screen is divided into two sections:

The Sidebar (left panel)

- MONITORING: Overview (live charts), Devices, Alerts

At the very top of the sidebar, you will see your name, email, and a colored role badge (ADMIN, TECHNICIAN, or VIEWER). At the bottom is the Sign Out button.

The Main Content Area (right panel)

Step-by-step: Navigating Between Pages

1. Look at the sidebar on the left side of the screen.

TIP: If you bookmark a page and cannot access it later (session expired), return to <http://localhost:8501> and log in again. Your session token is held in browser memory and does not persist across browser restarts.

Step-by-step: Signing Out

1. Scroll to the bottom of the sidebar.

IMPORTANT: Always sign out when you are done, especially on shared computers. Leaving an active session open on a shared machine is a security risk.

Practical Example: First Login as a New Employee

You are Sarah, a new junior technician. It is your first day.

1. Your manager tells you your credentials: sarah@company.com / Welcome123!

Chapter 3: Your Role and What You Can Access

What it is

The RMM system uses Role-Based Access Control (RBAC). This means different users see different options and have different levels of permission depending on their assigned role. There are three roles: admin, technician, and viewer.

Who uses it

Everyone should understand this chapter. Admins need it to manage other users. New staff need it to understand what they can and cannot do.

The Three Roles Explained

Viewer

Technician

Technicians cannot access the Admin page, manage other users, or configure billing.

Administrator

IMPORTANT: The Admin page is hard-restricted to admin role only. If you navigate to Admin without the admin role, you will see an error message and the page will stop loading.

How Roles Are Displayed

Your role is shown in the sidebar as a colored badge:

What Happens If You Try to Access Something Outside Your Role

If you are a viewer or technician and try to access the Admin page, you will see a red error message: "Admin access required. This page is restricted to admin users." The page will not load further.

For other restrictions (such as a viewer trying to create a ticket), the system enforces this at the API level -- the action button may appear but the API will return a permission error.

Practical Example: Understanding What You Can Do

Mark is a technician. On Monday morning, he notices a critical alert on the dashboard. He can:

PART II -- DAILY OPERATIONS

Chapter 4: The Dashboard -- Your Command Center

What it is

The Dashboard is the first page you land on after logging in. It is your at-a-glance health check for the entire system. There are actually two dashboard views:

1. The Home Page (app.py) -- shows the five key stat cards plus a navigation hint. This is what you see immediately after login.

Who uses it

Everyone. The dashboard is the starting point for all roles.

The Five Stat Cards

Both the home page and the Overview page display five stat cards across the top of the screen. These are live numbers pulled from the API:

| Card | What it shows |

The numbers update each time you load or refresh the page. They do not auto-refresh -- you must manually reload the page (press F5 or click your browser's refresh button) to see updated numbers.

TIP: The colored accent under each card tells you the health status. Green means normal, amber means caution, red means action required.

The Overview Page

Click Overview in the sidebar to go to the full dashboard view. It has four additional sections below the stat cards:

Device Status Donut Chart (left panel)

Device Health Map (right panel)

Recent Alerts (bottom left)

Activity Feed (bottom right)

Step-by-step: Morning Health Check

This is the recommended routine to start your shift:

1. Log in at <http://localhost:8501>.

Practical Example: Client Has a Server Down

ACME Corp calls you at 9am saying their server is unresponsive. Here is exactly what you do:

1. Open the Dashboard Overview.

Tips and Gotchas

- The dashboard does not auto-refresh. Always press F5 before making decisions based on numbers you see.

Chapter 5: Managing Tickets

What it is

The Tickets page is the helpdesk ticketing system. Every support request, incident, or task should have a corresponding ticket. Tickets allow you to track work, communicate with team members, and maintain a history of what was done for each client.

Who uses it

IT support staff, technicians, and administrators. Viewers can see tickets but cannot create or update them.

Ticket Fields

Field	Description	Options
-------	-------------	---------

Step-by-step: Creating a New Ticket

1. Click Tickets in the sidebar under MANAGEMENT.

WARNING: You must have at least one customer created before you can create a ticket. If the customer dropdown shows "-- no customers --", go to the Customers page and add the customer first.

Step-by-step: Finding a Ticket

1. Click Tickets in the sidebar.

Step-by-step: Updating a Ticket Status

1. Find the ticket using the search/filter steps above.

Step-by-step: Adding a Comment to a Ticket

1. Find and expand the ticket.

TIP: Use comments to keep a running log of everything you do on a ticket. Future team members reading the ticket should be able to understand the full history from comments alone.

Ticket Status Colors

- open -- red badge (needs attention)

Priority Colors

- critical -- red

Practical Example: Client Calls With a Complaint

It is 2pm. TechCorp Ltd calls and says their accounting software is running very slowly. Here is exactly what you do:

1. Go to Tickets in the sidebar.

Tips and Gotchas

- There is no automatic email notification to clients when a ticket is created or updated. Follow up with clients manually. (Alert rule triggers do send email notifications to configured addresses -- see Chapter 8.)

Chapter 6: Working with Customers

What it is

The Customers page manages the client organizations that own the devices you support. Every device belongs to a customer. Every ticket must be associated with a customer. Customer records hold contact information and support tier details.

Who uses it

IT support staff, technicians, and administrators.

Customer Fields

Field	Description	Options

Step-by-step: Creating a New Customer

1. Click Customers in the sidebar under MANAGEMENT.

Step-by-step: Editing a Customer

1. On the Customers page, find the customer in the list. Customers are displayed as expandable cards.

Step-by-step: Viewing Customer Details

1. On the Customers page, click any customer card to expand it.

Practical Example: Onboarding a New Client

You have just signed a new client: Greenway Manufacturing Ltd. Here is how to set them up:

1. Go to Customers in the sidebar.

Tips and Gotchas

- Customer names must be unique. If you try to create a customer with a name that already exists, the API will return an error.

Chapter 7: Devices -- Monitoring Your Fleet

What it is

The Devices page shows every registered device -- every computer, server, or workstation that has the RMM agent installed and has checked in with the system. This is the most information-dense page in the system.

Who uses it

All roles. Technicians and administrators interact most frequently with this page.

What You See on the Devices Page

The Devices page shows a list or table of all registered devices. For each device, you see:

| Column | Description |

Understanding Online vs Offline

A device is considered online if its agent has sent a heartbeat within the last few minutes. If no heartbeat has been received -- because the machine is powered off, the agent crashed, or there is a network issue -- the device is marked offline.

IMPORTANT: An offline device is not necessarily broken. It could simply be powered off outside business hours, or be on a network that is currently unreachable. Always check the Last Seen timestamp to understand how long the device has been offline.

Step-by-step: Viewing a Device's Detailed Metrics

1. Click Devices in the sidebar.

Status Indicators

The green or grey dot next to each device is the most important visual indicator on this page:

- Green dot with glow: Device is online and reporting normally.

On the Overview dashboard, devices are also categorized as:

Step-by-step: Investigating a Device After an Alert

You have received an alert that ACME-SRV01 has high CPU. Here is what to do:

1. Go to Devices in the sidebar.

Practical Example: New Employee Finding Their Assigned Devices

James is a new technician assigned to three small business clients. On his first day, he wants to see all their devices.

1. James goes to Customers in the sidebar.

TIP: If a device has not been seen for several hours during business hours, proactively contact the customer to verify the machine is powered on and network is working. Do not wait for the customer to call you.

Tips and Gotchas

- Last Seen time is shown in your local timezone. If your system time is wrong, the timestamps may appear incorrect.

Chapter 8: Alerts -- Staying Ahead of Problems

What it is

The Alerts page is where you manage all active system notifications. An alert is generated when a device's metrics cross a threshold you have defined -- for example, CPU usage above 90% for 5 minutes, or a disk at 95% capacity. Alerts are your early warning system.

The page has two tabs: Active Alerts and Alert Rules.

Who uses it

All roles can view alerts. Technicians and administrators can acknowledge, resolve, and create alert rules.

The Active Alerts Tab

Four stat cards at the top:

The severity filter dropdown lets you show only critical, warning, or info alerts. Select "All" to see everything.

The alert list shows each alert as an expandable row. The title shows the first 90 characters of the alert message. When expanded, you see:

- A colored severity bar on the left (red = critical, amber = warning, blue = info)

Alert Severity Levels

Severity	Color	Meaning
Critical	Red	Immediate action required
Warning	Amber	Action required soon
Info	Blue	Informational message

Alert Status

| Status | Meaning |

Step-by-step: Acknowledging an Alert

Acknowledging an alert means "I have seen this and I am dealing with it." It does not mean the problem is fixed -- it just stops the alert from appearing as new/unseen.

1. Click Alerts in the sidebar.

Step-by-step: Resolving an Alert

Resolving means the problem has been fixed. You should only resolve an alert after you have actually addressed the underlying issue.

1. Find and expand the alert.

The Alert Rules Tab

Alert rules define when alerts are created. Without rules, no alerts will ever be generated regardless of how bad device metrics become.

Existing rules are shown as expandable cards. Each shows:

Each rule has a Deactivate/Activate toggle and a Delete button.

Step-by-step: Creating an Alert Rule

1. Click Alerts in the sidebar.

Practical Example: Alert Fires at 2am

It is 2am. The on-call technician, Alex, receives an email notification: "CRITICAL: Disk usage on CORP-SRV02 is at 97%."

Here is exactly what Alex does:

1. Alex opens the system on their phone or laptop: <http://localhost:8501>

Step-by-step: Creating an Alert Rule for Disk Space

Here is the recommended disk space alert rule to set up for every environment:

1. Go to Alerts -> Alert Rules tab.

Repeat for a warning rule at 75%:

Tips and Gotchas

- If you get flooded with repeated alerts for the same device, check the cooldown setting. A cooldown of 5 minutes means a rule can fire 12 times per hour for a single device.

Chapter 9: App Center -- Software Inventory

What it is

The App Center (Software Inventory) page shows you all the software installed on a selected device. This is

useful for auditing what is running on client machines, identifying unauthorized software, checking software versions, and preparing for software patch management.

Who uses it

IT support staff and technicians primarily. Also useful for administrators conducting software audits.

What You See

The App Center page has a device selector at the top. Once you select a device:

- A summary shows how many software packages are installed on that device

A Check for Updates button is visible. In the current version, clicking it shows an informational message (full update functionality is part of Phase 6 and Software Patches).

Step-by-step: Viewing Installed Software on a Device

1. Click App Center in the sidebar (you may need to scroll down to find it under one of the navigation groups -- it may be listed as "App Center" or accessible via a sidebar link).

NOTE: Software inventory is collected by the agent every 6 hours (configurable via software_interval in the agent's config.ini). The data you see reflects the last software scan, not necessarily the current state. If software was installed in the last 6 hours, it may not appear yet.

Step-by-step: Checking for a Specific Application

1. Open App Center and select the device.

Practical Example: Software Audit for Compliance

Your manager asks you to confirm that all devices for DataSafe Ltd have Adobe Acrobat version 2024 or later.

1. Go to Customers and note all devices associated with DataSafe Ltd.

Tips and Gotchas

- The software list can be very long on a busy Windows machine (100+ applications). Use the search filter to find what you need quickly.

Chapter 10: Network Discovery

What it is

Network Discovery scans the local network and reports all hosts it can find -- computers, servers, network devices, printers. This is useful for discovering unregistered devices, auditing network exposure, or identifying what is connected to a client's network.

Who uses it

Technicians and administrators.

What the Scan Returns

After running a network scan, the results table shows:

Column	Description
--------	-------------

Step-by-step: Running a Network Scan

1. Click Network Discovery in the sidebar (under TOOLS).

NOTE: Network Discovery scans the local network segment that the RMM server is on. It does not scan remote client networks unless a scan agent is deployed on-site at those networks.

WARNING: Running a network scan on a network that monitors for intrusion detection may trigger alerts. Check with the client before running scans on sensitive networks.

Practical Example: Discovering Unauthorized Devices

A client suspects someone has connected an unauthorized device to their office network. You run Network Discovery and compare the results against the list of known devices registered in the RMM system. Any IP addresses or MAC addresses not matching a known device warrant investigation.

Tips and Gotchas

- Results are session-persisted, meaning they will remain on screen during your session but will not be saved to the database. Take a screenshot or note key findings if you need to refer to them later.

Chapter 11: Reports

What it is

The Reports page lets you generate formal reports about your managed environment. Reports are useful for client meetings, monthly reviews, compliance audits, and billing. Reports can be generated as PDF or Excel files.

Who uses it

Administrators and senior technicians.

Report Templates Available

Template	What it covers
----------	----------------

Step-by-step: Generating a Report

1. Click Reports in the sidebar (under BUSINESS).

Step-by-step: Viewing Report History

1. Click Reports in the sidebar.

Practical Example: Monthly Client Report

At the end of each month, you send each client a summary of their device health and any incidents. Here is the process:

1. Go to Reports -> Generate tab.

Tips and Gotchas

- If no data exists for the selected date range, the report may be empty or show "No data". Verify that devices were active and reporting during that period.

Chapter 12: Billing

What it is

The Billing page allows administrators to generate invoices for clients based on the number of devices managed during a billing period. It also tracks invoice status: draft, sent, or paid.

Who uses it

Administrators. Technicians may view billing but typically do not create invoices.

Billing Fields

Field	Description
-------	-------------

Step-by-step: Generating an Invoice

1. Click Billing in the sidebar (under BUSINESS).

Invoice Status

Status	Meaning
--------	---------

Step-by-step: Updating Invoice Status

1. Go to Billing in the sidebar.

Billing Summary Metrics

At the top of the Billing page you will see summary cards showing:

Practical Example: End-of-Month Billing Run

1. At the end of every month, go to Billing.

Tips and Gotchas

- Device count is entered manually. Always verify the actual number of managed devices before creating an invoice to avoid billing errors.

PART III -- ADVANCED OPERATIONS

Chapter 13: Administration

What it is

The Admin page is restricted to admin role users only. It provides three tabs: System Info, Audit Log, and Users. This is where you check system health, review all user actions, and manage user accounts.

Who uses it

Administrators only. Attempting to access this page as a technician or viewer shows an error.

Tab 1: System Info

The System Info tab shows three cards:

Current User card:

System card:

Services card:

NOTE: PostgreSQL and Redis status dots show amber with the address rather than a live probe. This is the configured address, not necessarily a live connection test. If you need to verify database connectivity, check the Flask API logs.

Tab 2: Audit Log

The Audit Log records every state-changing action in the system. This includes:

| Action Type | Color | Examples |

The audit log shows:

Filtering the Audit Log:

Step-by-step: Investigating a Suspicious Action

1. Go to Admin in the sidebar.

Tab 3: Users

The Users tab shows a table of all registered users in the system. Columns:

User management actions are available directly in the Users tab. You can:

Step-by-step: Deactivating a Departed Employee

When a team member leaves, their account must be deactivated promptly to prevent unauthorized access.

1. Go to Admin -> Users tab.

WARNING: JWT tokens have an expiry time. Even after deactivating an account, if the user has a valid token they may still have access until that token expires. Force-expire tokens by rotating the JWT secret key if immediate logout is required.

Practical Example: Monthly Admin Review

On the first Monday of each month, the administrator performs a security review:

1. Go to Admin -> Audit Log.

Tips and Gotchas

- The Audit Log is your accountability trail. Every admin action is logged with IP and user. Do not attempt to hide actions -- they are always recorded.

Chapter 14: Automation Profiles

What it is

Automation Profiles let you define a bundle of maintenance tasks that run automatically on a schedule -- daily, weekly, monthly, or on demand. Each profile combines OS patching, software patching, disk maintenance, and cleanup tasks into a single workflow that can be run across many devices with no manual intervention.

Who uses it

Administrators and senior technicians.

Understanding the Profile Structure

An automation profile has:

- Name and status: The profile's identifier and whether it is active or inactive.

| Column | Tasks Covered |

There is also a Scripts section where you can attach custom scripts to run as part of the profile.

The Profile List Tab

The Profile List tab shows all existing profiles as cards. Each card displays:

Clicking Run Now immediately queues the profile's tasks on all assigned devices.

Step-by-step: Creating an Automation Profile

1. Click Automation in the sidebar (under MANAGEMENT).

Schedule setup:

Task configuration:

12. Under SOFTWARE PATCH MANAGEMENT:

13. Under DISK MANAGEMENT:

14. Under MAINTENANCE:

15. In the SCRIPTS section, you can attach any scripts from the library to run after maintenance.

16. Click Save Profile.

Step-by-step: Running a Profile Immediately

1. Go to Automation -> Profile List tab.

Practical Example: Setting Up Weekly Patching for a New Client

Greenway Manufacturing has just been onboarded. They need weekly security patches every Sunday night.

1. Go to Automation -> Create / Edit Profile.

Disabling and Deleting Profiles

To disable a profile without deleting it:

To delete a profile:

WARNING: Deleting a profile cannot be undone. If you only want to pause it temporarily, disable it instead.

Tips and Gotchas

- Rebooting is part of the Maintenance column but is a high-impact action. Always coordinate with clients before enabling the Reboot checkbox in any automated profile.

Chapter 15: OS Patch Management

What it is

The OS Patches page manages Windows Update deployment across all your managed devices. You can see which patches are pending approval, approve batches of patches, review patch history, and configure policies.

Who uses it

Technicians and administrators.

The Four Stat Cards

| Card | Description |

Compliance color coding:

The Pending Patches Tab

This tab shows all patches waiting for your approval. Each patch shows:

Approving patches:

NOTE: Approving a patch does not immediately install it. The agent on the target device will receive the approval on its next heartbeat cycle and then proceed with installation.

The Patch History Tab

This tab shows all patches across all statuses -- pending, approved, deployed, and failed. It is displayed as a structured table with columns:

| Column | Description |

Use this tab for compliance audits and to verify that specific patches have been applied.

The Policies Tab

The Policies tab links to Automation Profiles for granular patch policy configuration. Patch policies -- including auto-approval rules, maintenance windows, and exclusions -- are managed through Automation Profiles (Chapter 14).

Step-by-step: Approving This Week's Security Patches

1. Go to OS Patches in the sidebar.

Practical Example: Checking Patch Compliance for a Client

The client DataSafe Ltd asks you to confirm that all their servers have the latest security patches. Here is how:

1. Go to OS Patches -> Patch History tab.

Tips and Gotchas

- Patch deployment requires the agent to be online. Devices that are offline will not receive patches until they reconnect and the agent checks in.

Chapter 16: Software Patches

What it is

The Software Patches page manages third-party software updates on individual devices -- applications like Chrome, Firefox, 7-Zip, VLC, and others that can be updated via winget or chocolatey package managers.

Who uses it

Technicians and administrators.

Page Layout

The page has a two-column layout:

Left column:

Right column:

Step-by-step: Checking for Software Updates on a Device

1. Click Software Patches in the sidebar (under PATCHING).

NOTE: Only online devices appear in the Software Patches device selector. If a device is offline, it cannot be patched remotely until it comes back online.

Understanding the Software Table

The right column shows the software inventory for the selected device. Each row shows the package name, version, and publisher. For packages detected as having available updates, this may be highlighted.

Practical Example: Updating Chrome on a Device

1. Go to Software Patches.

Tips and Gotchas

- Software patching via winget/chocolatey requires that these package managers are available on the device. Most modern Windows 10/11 systems have winget by default.

Chapter 17: Disk Management

What it is

Disk Management gives you a visual and actionable view of disk usage across any selected device. It shows gauge charts for each disk drive and provides action buttons to perform disk maintenance remotely.

Who uses it

Technicians and administrators.

What You See

After selecting a device, the page shows:

Gauge Charts (up to 4 drives):

Summary Table:

Action Buttons:

These actions are queued via the agent and executed remotely.

Step-by-step: Investigating High Disk Usage

1. Click Disk Management in the sidebar (under TOOLS).

Practical Example: Emergency Disk Cleanup

Alert fires at 3pm: RETAIL-PC05 disk usage at 93%. The machine cannot be rebooted during business hours.

1. Go to Disk Management.

Tips and Gotchas

- Do not defragment SSDs. Defragmentation wears out SSD cells and provides no performance benefit. Only run Defragment on traditional spinning hard drives (HDDs).

Chapter 18: Maintenance Actions

What it is

The Maintenance page lets you perform direct remote actions on a selected online device: reboot, shutdown, create restore points, delete temp files, clear browser history, and run a disk check. It also shows a log of recent maintenance runs.

Who uses it

Technicians and administrators.

Critical Safety Features

Only online devices are shown. The device selector only lists devices that are currently online. If a device is offline, it will not appear.

Confirmation checkbox required. Before any action button becomes effective, you must check the confirmation box: "I confirm this action on the selected device." If you click any action button without checking this box, you get a warning message and nothing happens.

IMPORTANT: Reboot and Shutdown are live commands -- they execute immediately on the device. Rebooting a server in use will disconnect all connected users. Shutting down a server will take it completely offline. Always confirm with the client before executing these actions.

The Device Info Card

After selecting a device, a card shows:

This card helps you confirm you have selected the correct device before taking any action.

Available Actions

Button	Effect	Risk Level
Reboot	Reboots the device	High
Shutdown	Shuts down the device	High
Create Restore Point	Creates a restore point	Low
Delete Temp Files	Deletes temporary files	Low
Clear Browser History	Clears browser history	Low
Check Disk	Checks disk for errors	Low

NOTE: All six actions are fully functional. Create Restore Point, Delete Temp Files, Clear Browser History, and Check Disk are queued as agent tasks -- the agent picks them up on its next poll cycle (within 60 seconds). Reboot and Shutdown execute via the same mechanism.

Step-by-step: Rebooting a Device

1. Click Maintenance in the sidebar (under TOOLS).

Step-by-step: Checking Recent Maintenance Runs

1. Scroll to the bottom of the Maintenance page (after selecting a device).

Practical Example: Coordinated Reboot of a Server After Patching

After an OS patch has been applied to CORP-SRV01, a reboot is required to complete installation. It is 8pm (after hours).

1. Go to Maintenance.

Tips and Gotchas

- The confirmation checkbox state is per device per session. If you switch devices, the checkbox resets.

Chapter 19: Scripts -- Running and Writing

What it is

The Scripts page is one of the most powerful features of the RMM system. It lets you run automated scripts on managed devices from a central interface -- no need to remote desktop into machines individually. You can run built-in scripts from the library, upload custom scripts, and review the execution history of all past runs.

Who uses it

Technicians and administrators for day-to-day use. Developers for creating and maintaining the script library.

Supported Script Types

| Type | Badge Color | Language | Typical Use |

The Three Tabs

Library Tab

Use the search field at the top to filter scripts by name or description.

Upload Tab

Click Upload Script to save it to the library.

Run History Tab

Expanding a run entry shows:

Step-by-step: Running a Script on a Single Device

1. Click Scripts in the sidebar (under TOOLS).

Step-by-step: Running a Script on Multiple Devices at Once

1. Follow steps 1-4 above.

Practical Example: Running a Cleanup Script on 10 Devices at Once

Your manager asks you to run the "Clear Temp Files" script on all 10 devices belonging to QuickPrint Co during off-hours.

1. Go to Scripts -> Library tab.

Step-by-step: Uploading a New Script

1. Click Scripts in the sidebar.

Reading Script Output

When a script completes, the output is captured by the agent and sent back to the API. To read it:

1. Go to Run History tab.

A successful script (exit code 0) with output in stdout is the expected pattern. If you see output in stderr, investigate even if exit code is 0 -- the script may have encountered non-fatal warnings.

Tips and Gotchas

- Only online devices appear in the device multi-select. If a target device is not shown, it is offline.

PART IV -- TECHNICAL REFERENCE

Chapter 20: System Architecture Overview

Component Map



Service Startup Order

Services must be started in this order:

1. PostgreSQL -- must be running before anything else. On Windows with Memurai: ensure the service is running in Windows Services.

Directory Structure

```
RemoteManagementSystem/
```

Authentication Flow

1. User submits email + password to dashboard login form.

Agent Registration Flow

1. Agent reads config.ini on startup.

Database Models Overview

Model	Table	Key Fields

Chapter 21: Script Writing Guide

Overview

Scripts are the automation workhorses of the RMM system. You can write scripts in PowerShell (ps1), Windows Batch (bat), Python (py), or Shell (sh). The agent executes them on the managed device and captures stdout and stderr output, which is sent back to the RMM and displayed in Run History.

PowerShell (ps1) -- Recommended for Windows

PowerShell is the preferred language for Windows automation. It has access to .NET, WMI, registry, COM objects, and all Windows management APIs.

Basic template:

```
try {
    Write-Output "=== System Information ==="
```

Key conventions for PS1 scripts:

Example: Get 10 Largest Files on C:

```
try {
    Write-Output "Top 10 largest files on C:"
```

Example: Clear Temp Files:

```
$tempPaths = @(
$freed = 0
$freedMB = [math]::Round($freed / 1MB, 1)
```

Windows Batch (bat)

Batch scripts are simpler and suitable for basic command execution. Less preferred than PowerShell but useful for legacy compatibility.

Template:

```
net start
```

Key conventions for BAT scripts:

Python (py)

Python scripts are useful for complex logic, data processing, or tasks that need libraries. The agent runs these using the Python interpreter.

Template:

```
import sys
def main():
if __name__ == "__main__":
```

Key conventions for Python scripts:

Shell (sh)

Shell scripts run on Linux/macOS agents. Use for non-Windows managed devices.

Template:

```
df -h
```

Best Practices for All Script Types

1. Always include a description -- future team members will thank you.

Chapter 22: Alert Rules -- Writing Effective Rules

Overview

Alert rules define the conditions under which the system automatically generates an alert. Each rule monitors one metric on one comparison operator with one threshold. Multiple rules combine to give you comprehensive coverage.

Rule Anatomy

Rule: "Critical CPU Alert"

When any device's CPU metric exceeds 90%, a critical alert is created. It will not fire again for the same device for 15 minutes (cooldown). A ticket is auto-created.

Available Metrics

Metric	What It Measures	Threshold	Unit
--------	------------------	-----------	------

Operators

Operator	Symbol	Use When
----------	--------	----------

Recommended Rule Set

For a standard managed environment, implement these rules as a baseline:

1. Critical CPU Alert
2. High CPU Warning
3. Critical RAM Alert
4. High RAM Warning
5. Critical Disk Alert
6. Disk Warning
7. Device Offline Alert
8. Low Battery Warning (for laptops)

Cooldown Tuning

The cooldown period is critical to avoid alert fatigue. If a device's CPU is consistently at 95%, and your cooldown is 5 minutes, you will receive 12 alerts per hour for that one device. This drowns out other alerts and causes fatigue.

Recommended cooldowns by severity:

Severity	Recommended Cooldown
----------	----------------------

For the "offline" metric, use a longer cooldown (60+ minutes) because network fluctuations can cause brief disconnects that are not real outages.

Auto-Ticket Policy

Use auto-create ticket conservatively:

An environment with too many auto-created tickets becomes noisy and technicians start ignoring them.

Chapter 23: Automation Profile Design

Design Principles

A well-designed automation profile answers these questions:

Profile Templates

Template 1: Nightly Security Maintenance (Standard)

Use for: all standard client devices, every night.

Template 2: Weekend Full Maintenance

Use for: servers and workstations where a weekly full maintenance is acceptable.

Template 3: Monthly Patch Tuesday

Use for: enterprise clients who prefer controlled monthly patching aligned with Microsoft's Patch Tuesday.

Avoiding Common Automation Mistakes

1. Do not schedule reboots on devices that are always in use. Check with clients before enabling Reboot in any profile.
2. Test on one device before fleet-wide rollout. Create the profile, set it to manual (on_demand), run it on one test device, and verify before activating the schedule.
3. Exclude known problematic packages. If a software update consistently breaks an application, add it to the excluded list in the Software Patch Management column.
4. Stagger schedules across clients. If all clients run their profiles at exactly 2:00 AM, the API and database will be hit simultaneously. Spread profiles across different times (2:00, 2:30, 3:00, etc.).
5. Review run history. After each profile run, check the Maintenance run log for failed runs. A profile that consistently fails on the same device needs investigation.

Chapter 24: User Roles and Permissions Matrix

Full Permissions Matrix

| Feature / Action | Viewer | Technician | Admin |

Chapter 25: Common Troubleshooting

Problem: Cannot log in -- "Invalid credentials"

Cause: Wrong email, wrong password, or account is deactivated.

Steps:

Problem: Dashboard shows "API error: [message]"

Cause: The Flask API at <http://localhost:5000> is not responding.

Steps:

Problem: Devices showing as offline when they should be online

Cause: The agent on the device has stopped running, the device is off, or there is a network issue.

Steps:

Problem: Agent won't register -- "Registration failed"

Cause: Wrong API URL in config.ini, wrong org_token, or API is not running.

Steps:

Problem: Scripts not running -- stuck in "queued" status

Cause: The agent is offline or not polling for tasks, or Celery is not running.

Steps:

Problem: "No pending patches" but devices are clearly not updated

Cause: The agent's patch discovery has not run yet, or the patch discovery feature is not yet fully implemented.

Steps:

Problem: Automation profile runs show as "failed"

Cause: A task in the profile encountered an error during execution.

Steps:

Problem: Cannot access Admin page -- "Admin access required"

Cause: Your account has the technician or viewer role, not admin.

Steps:

Problem: Dashboard is blank or shows error after navigating

Cause: JWT token has expired or is invalid.

Steps:

Starting All Services -- Quick Reference

```
# 1. Verify PostgreSQL is running (check Windows Services)
```

```
# 2. Start Redis/Memurai (Windows service or standalone)
```

```
# 3. Start Flask API
```

```
# 4. In a new terminal -- Start Celery Worker
```

```
# 5. In a new terminal -- Start Celery Beat
```

```
# 6. In a new terminal -- Start Streamlit Dashboard
```

```
# 7. On each managed machine -- Start Agent (as Administrator)
```


APPENDIX A: GLOSSARY

| Term | Definition |

APPENDIX B: QUICK REFERENCE CARDS

Quick Reference Card -- IT Support Staff

Your daily routine:

Creating a ticket:

Finding a device:

Acknowledging an alert:

Updating ticket status:

Sign out when done:

Quick Reference Card -- Technicians

Run a script on multiple devices:

Approve OS patches:

Reboot a device:

Create an alert rule:

Create an automation profile:

View disk health:

Check script output:

Alert severity colors:

Ticket status cycle:

Quick Reference Card -- Administrators

Access user management:

View audit log:

Check service health:

Create invoice:

Deactivate departed employee:

Service ports:

Restart all services (PowerShell):

Quick Reference Card -- Developers

Dashboard entry point: dashboard/app.py

Brand colors:

Authentication in dashboard pages:

API call pattern:

Agent heartbeat interval: 60 seconds (configurable: config.ini -> [agent] -> heartbeat_interval)

Script exit code convention: 0 = SUCCESS, non-zero = FAILED

Run tests:

End of RMM System Handover and User Guide -- Version 1.0

For questions about this guide, contact your system administrator or development team.